

Year 11 Term 4 Mathematics Extension 2 Exam, 2008

Question 1.	Marks
(a) Find the modulus and principal argument of $7 - 7i$.	2
(b) Simplify $\frac{5+2i}{3+4i}$ in the form $a+ib$ where $a, b \in R$.	2
(c) Sketch the locus on the Argand diagram : $ z + 2 - 2i = 2$.	3
(d) A root of $x^2 - 3x + b = 0$ is $2 - 4i$ where b is a complex number. Find the other root.	1
(e) (i) Find the three cube roots of -1 in the form $a+ib$ where $a, b \in R$.	3
(ii) Show accurately on the Argand diagram the points P_1, P_2 and P_3 representing the complex cube roots of -1 .	1
(f) The point P representing z in the Argand diagram satisfies the equation : $ z - 1 - 5i = z + 2 - 3i $.	2
(i) Find the Cartesian equation of the locus of the point P .	
(ii) Find the minimum value of $ z $.	1
 Question 2.	
(a) $OABC$ is a square where $\vec{OA}$ is represented by the vector $2 + 3i$. Find the vectors representing $\vec{OB}$ and $\vec{OC}$.	4
(b) (i) Find the square roots of $5 - 12i$.	2
(ii) Hence solve $x^2 - (8 - 6i)x + 2 - 12i = 0$.	2
(c) (i) Write in Modulus Argument form : $3\sqrt{3} + 3i$.	2
(ii) Hence show that $3\sqrt{3} + 3i$ is a root of $x^4 - 36x^2 + 1296 = 0$.	3
(iii) Deduce a quadratic factor of $x^4 - 36x^2 + 1296$ with real coefficients giving reasons.	2

Question 3.

(a) 2

(i) If $z = r(\cos \theta + i \sin \theta)$ and $|z| = 1$ show $\cos n\theta = \frac{z^n + \frac{1}{z^n}}{2}$
and $\sin n\theta = \frac{z^n - \frac{1}{z^n}}{2i}$ for

n integer.

(ii) Find an expression for $\sin n\theta \sin 2\theta$ in terms of z , and deduce that 2

$$\sin n\theta \sin 2\theta = \frac{1}{2} \{ \cos(n-2)\theta - \cos(n+2)\theta \}.$$

(ii) Evaluate $\int_0^{2\pi} \sin n\theta \sin 2\theta d\theta$ if n is an integer. 4

(b) In the Argand diagram the points A, B and C represent the triangle ABC .

The vectors $\vec{AB}$ and $\vec{AC}$ are represented by z_1 and z_2 respectively with

$$\text{Arg } z_1 = \alpha, \text{Arg } z_2 = \beta \text{ and } \angle BAC = \theta.$$

(i) Draw the triangle ABC in the Argand diagram with $\alpha > \beta > 0$. 1

(ii) Write z_1 and z_2 in Mod Arg form. 2

(iii) Show that the Area A of triangle ABC is given by : 2

$$A = \left| \frac{1}{2} \text{Im} \left(z_1 \bar{z}_2 \right) \right|.$$

(iv) Hence find the area of $\triangle ABC$ if the coordinates in the x - y plane of A, B and C are 2

$(3, 2), (7, 5)$ and $(9, 4)$ respectively.

Question 4.

(a) Graph the intersection of the regions defined by : 5

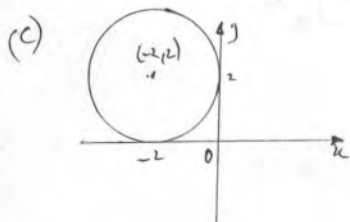
$$z \bar{z} \geq 9, z + \bar{z} \leq 8 \text{ and } 0 < \text{Arg } z < \frac{\pi}{4}.$$

- (b) Let α be the complex root of $z^7 = 1$ with the smallest positive argument.
 Let $\mathcal{G} = \alpha + \alpha^2 + \alpha^4$ and $\delta = \alpha^3 + \alpha^5 + \alpha^6$.
- (i) Write α in Mod-Arg form. **1**
- (ii) Explain why $1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 = 0$ **1**
- (iii) Show $\mathcal{G} + \delta = -1$ and $\mathcal{G}\delta = 2$. **2**
- (iv) Write a quadratic equation in the variable t with roots $\mathcal{G}$ and δ . **1**
- (v) Solve this quadratic equation and deduce that $\mathcal{G} = -\frac{1}{2} + i\frac{\sqrt{7}}{2}$ **3**
 and **not** $-\frac{1}{2} - i\frac{\sqrt{7}}{2}$.
- (vi) Find the value of $\cos\frac{4\pi}{7} + \cos\frac{2\pi}{7} - \cos\frac{\pi}{7}$ **1**
- (vii) Deduce the value of $\sin\frac{4\pi}{7} + \sin\frac{2\pi}{7} - \sin\frac{\pi}{7}$. **1**

End of Exam

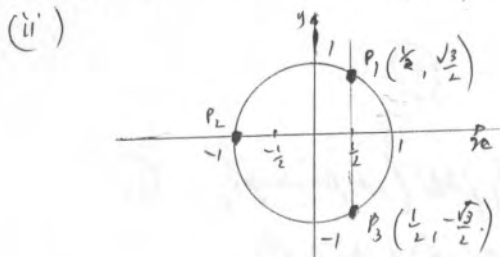
(a) $7-7i$ Modulus $|z| = 7\sqrt{2}$ units
Argument $= -\frac{\pi}{4}$

(b) $\frac{5+2i}{3+4i} = \frac{(5+2i)(3-4i)}{(3+4i)(3-4i)}$
 $= \frac{15-20i+6i+8}{3^2+4^2}$
 $= \frac{23-14i}{25}$
 $= \frac{23}{25} - \frac{14}{25}i$



(d) $z_1 = 2-4i$
 $z_1 + z_2 = 3$
 $\therefore$ other root $z_2 = 1+4i$

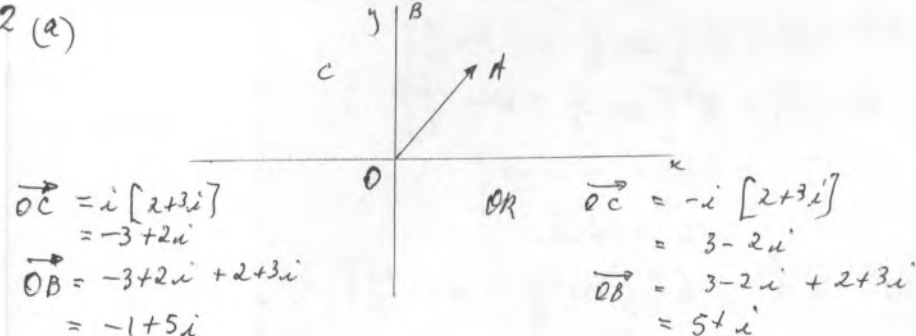
(e) (i) $z^3 = -1$
 $z = \cos\left(\frac{\pi+2\pi k}{3}\right) + i\sin\left(\frac{\pi+2\pi k}{3}\right)$ k integer
 $k=0$ $z_0 = \cos\frac{\pi}{3} + i\sin\frac{\pi}{3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i$
 $k=1$ $z_1 = \cos\pi + i\sin\pi = -1 + 0i$
 $k=2$ $z_2 = \cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3} = \frac{1}{2} - \frac{\sqrt{3}}{2}i$
 $\therefore$ roots of -1 are $\frac{1}{2} + \frac{\sqrt{3}}{2}i$, $-1 + 0i$, $\frac{1}{2} - \frac{\sqrt{3}}{2}i$



(f) (i) $(x-1)^2 + (y-5)^2 = (x+2)^2 + (y-3)^2$
 $-2x+1 -10y+25 = 4x+4y+9$
 $6x+4y-13=0$

(ii) $|z|_{\min} = \frac{13}{\sqrt{6^2+4^2}}$
 $= \frac{13}{\sqrt{52}}$
 $= \frac{\sqrt{13}}{2}$ units.

2 (a)



(b) $(x+iy)^2 = 5-12i$ $x, y \in \mathbb{R}$
 $x^2 - y^2 = 5$ (1)
 $2xy = -12$
 $xy = -6$ (2)
 $y = \frac{-6}{x}$

$\therefore x^2 - \frac{36}{x^2} = 5$

$x^4 - 5x^2 - 36 = 0$

$(x^2-9)(x^2+4) = 0$

$x^2 = 9$ or $x^2 = -4$

$x=3$ $y=-2$ No other $x \in \mathbb{R}$

$x=-3$ $y=2$

$\therefore$ square roots of $5-12i$ are $\pm(3-2i)$

(ii) $x^2 - (8-6i)x + 2-12i = 0$

$x = \frac{8-6i \pm \sqrt{(8-6i)^2 - 8+48i}}{2}$

$= \frac{8-6i \pm \sqrt{64-96i-36-8+48i}}{2}$

$= \frac{8-6i \pm \sqrt{20-48i}}{2}$

$= 4-3i \pm \sqrt{5-12i}$

$= 4-3i \pm (3-2i)$

$x = 7-5i$ or $1-i$

$$(i) \quad (3\sqrt{3}+3i)^2 = 6^2 \left[\cos \frac{2\pi}{6} + i \sin \frac{2\pi}{6} \right] \\ = 36 \left[\frac{1}{2} + i \frac{\sqrt{3}}{2} \right] \\ = 18 + 18\sqrt{3}i$$

$$(3\sqrt{3}+3i)^4 = 6^4 \left[\cos \frac{4\pi}{6} + i \sin \frac{4\pi}{6} \right] \\ = 1296 \left[-\frac{1}{2} + i \frac{\sqrt{3}}{2} \right] \\ = -648 + 648\sqrt{3}i$$

$$\therefore x^4 - 36x^2 + 1296 = -648 + 648\sqrt{3}i - 36(18 + 18\sqrt{3}i) + 1296 \\ = -648 + 648\sqrt{3}i - 648 - 648\sqrt{3}i + 1296 \\ = 0$$

$\therefore x = 3\sqrt{3} + 3i$ is a root of $x^4 - 36x^2 + 1296 = 0$

(ii) Another root is $x = 3\sqrt{3} - 3i$ [conjugate root with real coefficients]

$$\therefore \text{Quadratic factor } (x - 3\sqrt{3} - 3i)(x - 3\sqrt{3} + 3i) \\ = (x - 3\sqrt{3})^2 + 9 \\ = x^2 - 6\sqrt{3}x + 36 \quad \text{OR } x^2 + 6\sqrt{3}x + 36$$

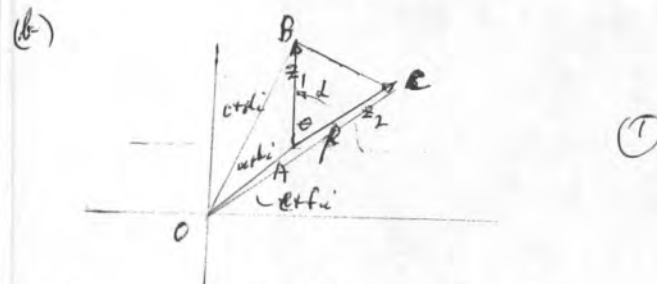
$$(i) \quad |z|=1 \quad \therefore z = \cos \theta + i \sin \theta \\ z^n = (\cos \theta + i \sin \theta)^n \\ z^n = \cos n\theta + i \sin n\theta \quad (1) \\ z^{-n} = \cos(-n\theta) + i \sin(-n\theta) \\ \frac{1}{z^n} = \cos n\theta - i \sin n\theta \quad (2) \\ \therefore z^n + \frac{1}{z^n} = 2 \cos n\theta \quad z^n - \frac{1}{z^n} = 2i \sin n\theta \\ \cos n\theta = \frac{z^n + \frac{1}{z^n}}{2} \quad \& \quad \sin n\theta = \frac{z^n - \frac{1}{z^n}}{2i}$$

$$\therefore \sin n\theta \sin 2\theta = \frac{z^n - \frac{1}{z^n}}{2i} \cdot \frac{z^2 - \frac{1}{z^2}}{2i} \\ = \frac{1}{4} \left[z^{n+2} - z^{n-2} - \frac{1}{z^{n-2}} + \frac{1}{z^{n+2}} \right] \\ = \frac{1}{4} \left[z^{n+2} + \frac{1}{z^{n+2}} - \left(z^{n-2} + \frac{1}{z^{n-2}} \right) \right]$$

$$\therefore \sin n\theta \sin 2\theta = \frac{1}{2} \left[\frac{z^{n+2} + \frac{1}{z^{n+2}}}{2} - \left(\frac{z^{n-2} + \frac{1}{z^{n-2}}}{2} \right) \right] \\ = \frac{1}{2} \left[\cos(n+2)\theta - \cos(n-2)\theta \right] \quad (1)$$

$$(i) \int_0^{2\pi} \sin n\theta \sin 2\theta d\theta = \frac{1}{2} \int_0^{2\pi} [\cos(n-2)\theta - \cos(n+2)\theta] d\theta \\ = \frac{1}{2} \left[\frac{\sin(n-2)\theta}{n-2} - \frac{\sin(n+2)\theta}{n+2} \right]_0^{2\pi} \\ = 0 \quad n \neq \pm 2 \quad (2)$$

$$\text{OR} \quad = \frac{1}{2} \int_0^{2\pi} 1 - \cos 4\theta d\theta \quad n=2 \\ = \frac{1}{2} \left[\theta - \frac{\sin 4\theta}{4} \right]_0^{2\pi} \\ = \pi \quad n=2 \\ \text{OR} = -\pi \quad n=-2 \quad (1)$$



$$(ii) \quad z_1 = |AB| \{ \cos \theta + i \sin \theta \} \\ z_2 = |AC| \{ \cos \phi + i \sin \phi \} \quad (1)$$

$$(iii) \quad \text{RHS} = \frac{1}{2} \text{Im} \left\{ |AB| \{ \cos \theta + i \sin \theta \} / |AC| \{ \cos \phi + i \sin \phi \} \right\} \\ = \frac{1}{2} \text{Im} \left\{ |AB|/|AC| \{ \cos \theta + i \sin \theta \} \{ \cos(-\phi) + i \sin(-\phi) \} \right\} \\ = \frac{1}{2} \text{Im} |AB|/|AC| \{ \cos(\theta-\phi) + i \sin(\theta-\phi) \} \quad (1)$$

$$= \frac{1}{2} |AB| |AC| \sin(\angle B)$$

$$= \frac{1}{2} |AB| |AC| \sin \theta \text{ where } \angle B = \theta$$

$$= \text{Area } \triangle ABC.$$

$$(iv) \quad z_1 = 4+3i$$

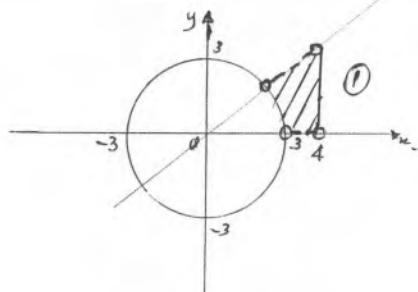
$$z_2 = 6+2i$$

$$\text{Area} = \frac{1}{2} \text{Im}(z_1 \bar{z}_2)$$

$$= \frac{1}{2} \text{Im}[(4+3i)(6-2i)]$$

$$= \frac{1}{2} [-8 + 18]$$

$$= 5 \text{ unit}^2.$$



$$\text{Q4} \quad z\bar{z} \geq 9$$

$$\text{or } x^2 + y^2 \geq 9 \quad (1)$$

$$z + \bar{z} \leq 8$$

$$2x \leq 8$$

$$x \leq 4 \quad (2)$$

$$0 < \text{Arg } z < \frac{\pi}{4} \quad (3)$$

$$(b) (i) \quad z^7 = 1 \Rightarrow z = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} \text{ smallest argument.}$$

$$\therefore z^7 - 1 = 0$$

$$(z-1)(z^6 + z^5 + z^4 + z^3 + z^2 + z + 1) = 0$$

$$\text{or } (z-1)(1+z+z^2+z^3+z^4+z^5+z^6) = 0$$

$$\therefore \text{If } z \text{ is complex then } z \neq 1$$

$$\text{hence } 1+z+z^2+z^3+z^4+z^5+z^6 = 0$$

$$(ii) \quad \theta = 1 + z^2 + z^4 \quad \delta = z^3 + z^5 + z^6$$

$$\theta + \delta = 1 + z^2 + z^3 + z^4 + z^5 + z^6$$

$$= -1$$

$$\text{And } \theta \delta = (1 + z^2 + z^4)(z^3 + z^5 + z^6)$$

$$= 1(z^3 + z^5 + z^6) + z^2(z^3 + z^5 + z^6) + z^4(z^3 + z^5 + z^6)$$

$$= z^3 + z^5 + z^6 + z^5 + z^7 + z^8 + z^7 + z^9 + z^{10}$$

$$= z^3 + z^5 + z^6 + z^5 + 1 + z + z^2 + z^3 + z^4 + z^5 + z^6$$

$$= 1 + z + z^2 + z^3 + z^4 + z^5 + z^6$$

$$= 1 + z + z^2 + z^3 + z^4 + z^5 + z^6$$

$$= 2 \times 1$$

$$= 2$$

$$(iv) \quad \theta + \delta = -\frac{b}{a} \quad \theta \delta = \frac{c}{a}$$

$$\therefore \frac{b}{a} = 1$$

$$\therefore \frac{c}{a} = 2$$

$$x^2 + \left(\frac{b}{a}\right)x + \frac{c}{a} = 0$$

$$x^2 + x + 2 = 0$$

$$(v) \quad x = \frac{-1 \pm \sqrt{1-8}}{2}$$

$$= \frac{-1 \pm \sqrt{-7}}{2} i$$

$$\text{Now } \theta = 1 + z^2 + z^4$$

$$= \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} + \left(\cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}\right)^2 + \left(\cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}\right)^4$$

$$= \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} + \cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7} + \cos \frac{8\pi}{7} + i \sin \frac{8\pi}{7}$$

$$= \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} + \cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7} - \cos \frac{\pi}{7} - i \sin \frac{\pi}{7}$$

$$= \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} - \cos \frac{\pi}{7} + i \left[\sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} - \sin \frac{\pi}{7} \right]$$

But $\sin \frac{\pi}{2} > \sin \frac{2\pi}{7} > \sin \frac{\pi}{7} > 0$

$\therefore \sin \frac{2\pi}{7} - \sin \frac{\pi}{7} > 0$

$\sin \frac{4\pi}{7} > 0$

$\therefore \sin \frac{4\pi}{7} + \sin \frac{2\pi}{7} - \sin \frac{\pi}{7} > 0$

$\operatorname{Im}(t) > 0.$

$\therefore \theta = -\frac{1}{2} + \frac{\sqrt{7}}{2}i$

(v1) $\cos \frac{4\pi}{7} + \cos \frac{2\pi}{7} - \cos \frac{\pi}{7} = -\frac{1}{2}$

(v4) $\sin \frac{4\pi}{7} + \sin \frac{2\pi}{7} - \sin \frac{\pi}{7} = \frac{\sqrt{7}}{2}.$